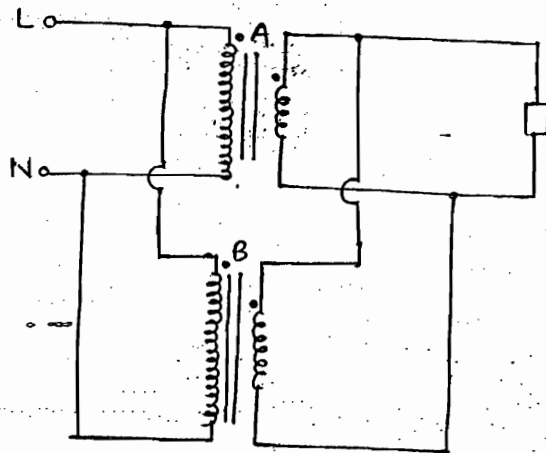


Parallel Operation OF Transformer

Condⁿ to be satisfied for parallel operation of Xmer $\rightarrow$

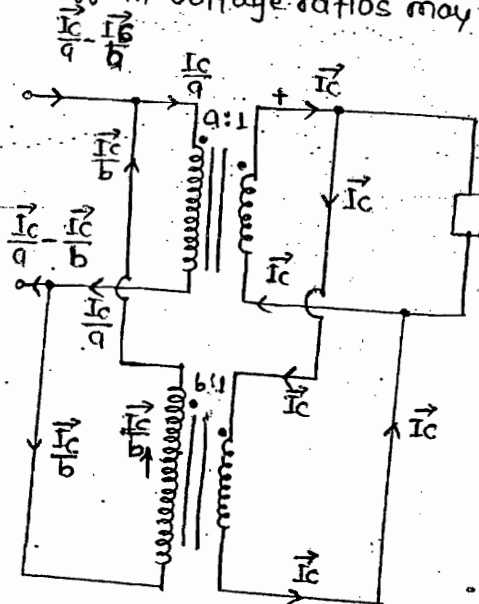
(A) For 1- ϕ TF & 3- ϕ TF $\rightarrow$

(a) Same polarity. i.e. the dots must be connected to dots. (must)



(b) Equal vol. Ratios (same vol. Rating) (must)

Note $\rightarrow$ A small diff. in voltage ratios may be permitted if unavoidable.



$$E_A > E_B$$

$$\frac{V_1}{a} > \frac{V_1}{b}$$

$$b > a$$

$$I_{cb} > I_{ca}$$

$$\frac{I_c}{a} > \frac{I_c}{b}$$

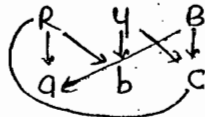
(c) Same pu impedance For proportional load sharing. (desirable)

$\downarrow$
Name plate z_{pu}

(d.) Same $\frac{X}{R}$ ratios. (i.e. same impedance angle) for same PF operation, as that of the load PF (desirable).

(B.) For 3- ϕ TF only $\rightarrow$

(e.) same phase sequence (MUST)



(f.) Zero phase diff between the corresponding line voltage.

This means that TF belonging to same phasor group may alone be parallel. (MUST)

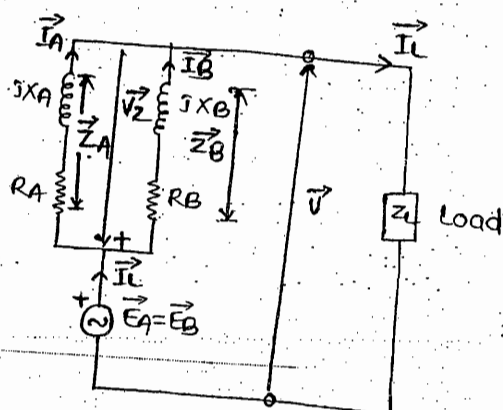
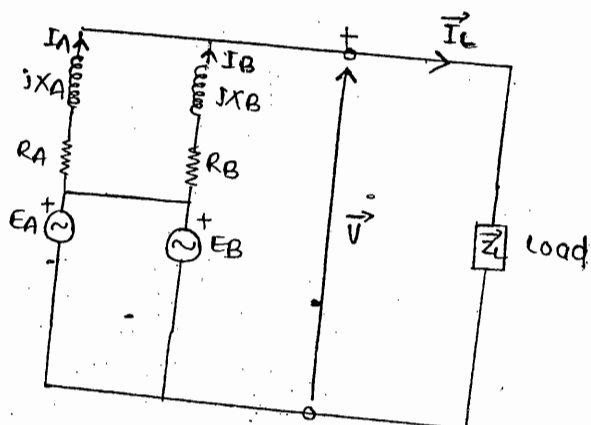
DY11 (a) Yy0 (b) Dd0 (c) Yy6 (d) Yd1

Because if we change the seq. of 1° then get 30° lead/lag.

LOAD SHARING

Equal Voltage Ratios

$$E_A = E_B$$



Two Xmer;

$$\vec{V}_Z = \vec{I}_A \vec{Z}_A = \vec{I}_B \vec{Z}_B = \vec{I}_L \times \vec{Z}_P$$

$$= \vec{I}_L \times \frac{\vec{Z}_A \vec{Z}_B}{\vec{Z}_A + \vec{Z}_B}$$

where; $Z_A \neq Z_B$ are in actual volts.

$$\vec{I}_A = \frac{Z_B}{Z_A + Z_B} \times \vec{I}_L \quad \& \quad \vec{I}_B = \frac{Z_A}{Z_A + Z_B} \times \vec{I}_L$$

However if the impedance are expressed in pu then there pu values should be converted on a common base to insure that the ohmic ratio remain unchanged.

for 'n' Xmers $\rightarrow$

$$\vec{V}_Z = \vec{I}_A \vec{Z}_A + \vec{I}_B \vec{Z}_B = \vec{I}_C \vec{Z}_C = \dots = \vec{I}_n \vec{Z}_n = \vec{I}_L \vec{Z}_p$$

$$\Rightarrow \vec{I}_A \vec{Z}_A = \vec{I}_L \vec{Z}_p$$

$$\Rightarrow \vec{I}_L = \frac{\vec{I}_A \vec{Z}_A}{\vec{Z}_p}$$

$$\Rightarrow \frac{\vec{I}_A \vec{Z}_A}{\left(\frac{1}{\vec{Z}_A} + \frac{1}{\vec{Z}_B} + \frac{1}{\vec{Z}_C} + \dots + \frac{1}{\vec{Z}_n}\right)} = \vec{I}_L$$

$$\vec{I}_A \vec{Z}_A = \frac{\vec{I}_L}{\left(\frac{1}{\vec{Z}_A} + \frac{1}{\vec{Z}_B} + \dots + \frac{1}{\vec{Z}_n}\right)}$$

$$\vec{I}_A = \frac{\vec{I}_L}{\vec{Z}_A \left(\frac{1}{\vec{Z}_A} + \frac{1}{\vec{Z}_B} + \dots + \frac{1}{\vec{Z}_n}\right)}$$

$$\boxed{\vec{I}_A = \frac{\vec{I}_L}{\left(\frac{\vec{Z}_A}{\vec{Z}_A} + \frac{\vec{Z}_A}{\vec{Z}_B} + \dots + \frac{\vec{Z}_A}{\vec{Z}_n}\right)}}$$

$$S_A = P_A + jQ_A = V_A I_A^*$$

$$S_A^* = P_A - jQ_A = V^* I_A$$

$$= V^* \frac{\vec{Z}_B}{\vec{Z}_A + \vec{Z}_B} \times \vec{I}_L$$

$$\boxed{S_A^* = \frac{\vec{Z}_B}{\vec{Z}_A + \vec{Z}_B} \times S_L^*}$$

Similarly;

$$\vec{S}_B = \vec{V} \cdot \vec{I}_B^*$$

$$S_B^* = V^* \vec{I}_B$$

$$\boxed{S_B^* = \frac{\vec{Z}_A}{\vec{Z}_A + \vec{Z}_B} \times S_L^*}$$

Per unit loading

$$\vec{I}_j \vec{Z}_j(\omega) = \text{Constant}$$

$$\vec{I}_j \propto \frac{1}{\vec{Z}_j(\omega)}$$

$$V \cdot \vec{I}_j \propto \frac{1}{\vec{Z}_j(\omega)}$$

$$S_j^* \propto \frac{1}{\vec{Z}_j(\omega)}$$

$$S_j^* \propto \frac{1}{\vec{Z}_j(\text{pu}) \times \vec{Z}_j(\text{Base})}$$

$$S_j^* \propto \frac{1}{\vec{Z}_j(\text{pu}) \times \left[\frac{V_{\text{rated}}^2}{S_{j(\text{rated})}} \right]}$$

$$S_j^* \propto \frac{S_{j(\text{rated})}}{\vec{Z}_j(\text{pu})}$$

$$\frac{S_j^*}{S_{j(\text{rated})}} \propto \frac{1}{\vec{Z}_j(\text{pu})}$$

$$S_j^*(\text{pu}) \propto \frac{1}{\vec{Z}_j(\text{pu})}$$

* This means that the TF with lowest pu impedance on its own base (i.e. name plate Z_{pu}) would have max^m pu loading & therefore would be the 1st to reach its FL.

For proportional load sharing

$$S_j \propto S_{j(\text{rated})}$$

$$\frac{S_j}{S_{j(\text{rated})}} = \text{constant}$$

$$S_j(\text{pu}) = \text{constant}$$

$$\text{Since } S_j(\text{pu}) \propto \frac{1}{\vec{Z}_j(\text{pu})}$$

$\therefore$ For proportional load sharing

$$\vec{Z}_j(\text{pu}) = \text{constant}$$

Que. → 2- ϕ Xmers rated 1000 kVA & 500 kVA respectively are connected in parallel on both HV & LV sides. They have equal voltage ratings of 11 kV/400V & their pu impedances are $0.02 + j0.07$ & $0.0455 + j0.0788$ pu respectively.

(A) How will the following loads will share
360 kVA at 0.9 PF lagging.

500 kW at UPF.

(B) What is the largest value of 0.8 PF lagging load that can be delivered by the parallel combination at the rated vol.

Determine the load shared & operating PF of the 2 TF under these condⁿ.

Solⁿ → $Z_A = 0.0728 \angle 74.05^\circ$ pu ; $Z_B = 0.091 \angle 60^\circ$ pu

Because the value of the Z_A is less, then it will go for FL1st

Selecting 1000 kVA as common base ; Z_A will same

$$(Z_B)_{\text{new}} = 0.091 \angle 60^\circ \times \frac{1000}{500}$$

$$(Z_B)_{\text{new}} = 0.182 \angle 60^\circ \text{ pu}$$

$$S_L = 360 \text{ kVA}$$

Part (A).

$$(i) S_L = 360 \angle -25.84^\circ \text{ pu kVA}$$

$$\vec{S}_A^* = \frac{(Z_B)_{\text{new}}}{Z_A + (Z_B)_{\text{new}}} \vec{S}_L^*$$

$$= \frac{0.182 \angle 60^\circ}{0.0728 \angle 74.05^\circ + 0.182 \angle 60^\circ} \times 360 \angle -25.84^\circ$$

$$= 258.73 \angle -29.84^\circ \text{ kVA}$$

i.e. 224.42 kW at 0.8674 PF lagging.

$$\text{For 2 TF only, } S_B = \vec{S}_L - \vec{S}_A$$

$$\& \text{ For 1 TF } S_B^* = \frac{Z_A}{Z_A + (Z_B)_{\text{new}}} \times \vec{S}_L^*$$

$$\vec{S}_B = 103.49 \angle 15.8^\circ \text{ kVA}$$

i.e. 99.58 kW at 0.9622 PF lag.

$$\text{QII } S_L = \frac{500}{10} \angle 0^\circ \text{ kVA}$$

$$S_A^* = \frac{Z_B(\text{new})}{Z_A + Z_B(\text{new})} \times 500 \angle 0^\circ$$

$$= 359.34 \angle -4^\circ \text{ kVA}$$

i.e. 358.47 kW at 0.9976 PF lag.

$$\text{For 2 Xmer only; } \vec{S}_B = \vec{S}_L - \vec{S}_A$$

$$\text{If TF use common formula } S_B^* = \frac{\vec{Z}_A}{\vec{Z}_A + \vec{Z}_B(\text{new})} \times S_L^*$$

$$\vec{S}_B = 143.74 \angle -10.04^\circ \text{ kVA}$$

i.e. 141.54 kW at 0.9847 PF lead.

(B) Since $S_{pu} \propto \frac{1}{Z_{pu}}$; TF A would reach FL; 1st

$$S_{pu} \propto \frac{1}{Z_{pu}}$$

$$Z_A < Z_B$$

$$\text{Then } \vec{S}_A = 1000 \angle 0^\circ \text{ kVA}$$

$$\text{Since } S_j^* \propto \frac{1}{Z_j(\Omega)}$$

$$\frac{S_B^*}{S_A^*} = \frac{Z_A(\Omega)}{Z_B(\Omega)} \quad \text{OR} \quad \frac{S_B^*}{S_A^*} = \frac{\vec{Z}_A}{\vec{Z}_B(\text{new})}$$

$$S_B^* = S_A^* \frac{Z_A}{Z_B(\text{new})}$$

$$S_L^* = S_A^* + S_B^*$$

$$= S_A^* + S_A^* \times \frac{\vec{Z}_A}{\vec{Z}_B(\text{new})}$$

$$= S_A^* \left[1 + \frac{\vec{Z}_A}{\vec{Z}_B(\text{new})} \right]$$

$$S_L^* = S_A^* [1.3914 \angle 4^\circ]$$

$$= 1000 \angle -\phi_A * 1.3914 \angle 4^\circ$$

$$S_L \angle -36.87^\circ = 1391.4 \angle 4^\circ - \phi_A \text{ kVA}$$

$$\max S_L = 1391.4 \text{ kVA}$$

$$\text{Equating angles} \Rightarrow -36.87^\circ = 4^\circ - \phi_A$$

$$\phi_A = 40.87^\circ$$

$$\text{i.e. } \vec{S}_A = 1000 \angle 40.87^\circ \text{ kVA}$$

$$P = 756.2 \text{ kW at } 0.7562 \text{ PF lag.}$$

$$\text{New } \vec{S}_B = S_L - S_A \text{ for 2 TF}$$

$$\text{Also in general } S_B^* = S_A^* \times \frac{\vec{Z}_A}{\vec{Z}_B(\text{new})}$$

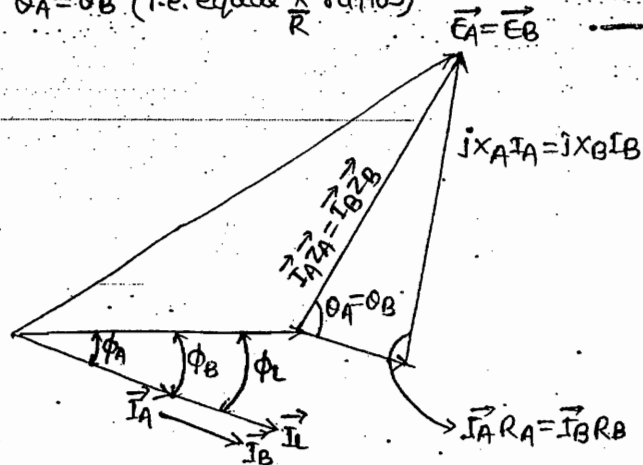
$$\vec{S}_B = 399.97 \angle 26.83^\circ \text{ kVA}$$

$$\text{i.e. } 356.92 \text{ kW at } 0.8923 \text{ PF lag.}$$

Phasor diagram →

$$E_A = E_B$$

$$\theta_A = \theta_B \text{ (i.e. equal } \frac{X}{R} \text{ ratios)}$$



$\theta_A > \theta_B$ (a unequal $\frac{x}{R}$ ratio)

The diagram shows a phasor system with terminal voltage $\vec{V}$ as the reference phasor. The induced EMF $\vec{E}_A = \vec{E}_B$ is the resultant of the field flux $\vec{\Phi}_A$ and the armature reaction flux $\vec{\Phi}_{AR}$. The armature current $\vec{I}_A$ is shown leading the terminal voltage $\vec{V}$ by an angle ϕ_A . The field flux $\vec{\Phi}_A$ leads $\vec{V}$ by $\phi_A - \phi_B$. The armature reaction flux $\vec{\Phi}_{AR}$ is in phase with $\vec{I}_A$. The terminal voltage $\vec{V}$ is the phasor sum of the induced EMF $\vec{E}_A$ and the voltage drops across the synchronous reactance $\vec{I}_A X_A$ and the synchronous resistance $\vec{I}_A R_A$. The angle between $\vec{V}$ and $\vec{E}_A$ is $\phi_A - \phi_B$.

$$E_A > E_B$$

$$= (\vec{I}_L \cdot \vec{Z}_L) + \vec{I}_A \vec{Z}_A$$

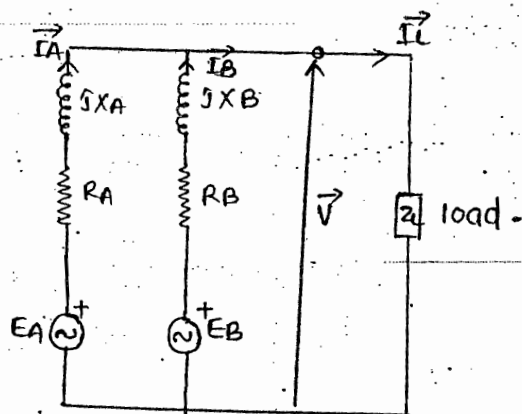
$$(\vec{Z}_A + \vec{Z}_B) \cdot \vec{I}_A + \vec{I}_B \cdot \vec{Z}_L = \vec{E}_A \quad \text{--- (i)}$$

Steps $\rightarrow Z_L \vec{I}_A + (Z_B + Z_L) \vec{I}_B = \vec{I}_B$ ——— (ii)

$$(ii) \vec{I}_L = \vec{I}_A + \vec{I}_B$$

(iii) $\vec{V} = I_L \vec{Z}_L$

(v) $\vec{S}_A = \vec{V} \cdot I_A^*$; $\vec{S}_B = \vec{V} \cdot I_B^*$



steps $\rightarrow$

$$(i) \quad \vec{V} = I_{SC} \vec{Z}_p$$

where $\vec{I}_{sc} = \sum_{j=1}^n \frac{E_j}{Z_j}$

$$\frac{1}{z_p} = \frac{1}{z_L} + \sum_{h=1}^n \frac{1}{z_j}$$

$$(2) \vec{I}_j = \frac{\vec{E}_j - \vec{V}}{\vec{Z}_j} \quad \text{where; } j=1, 2, \dots, n.$$

$$(3) \vec{S}_j = \vec{V} \cdot \vec{I}_j^* \quad \text{where; } j=1, 2, \dots, n$$

$$\vec{S}_L = \sum_{j=1}^n \vec{S}_j$$

Calculation check

$$\vec{S}_L = \vec{V} \cdot \vec{I}_L^*$$

$$\text{Also; } \vec{S}_L = \frac{(V)^2}{\vec{Z}_L^*}$$

$$= V \cdot \left(\frac{V}{\vec{Z}_L} \right)^*$$

$$= \frac{V \cdot V^*}{\vec{Z}_L^*}$$

$$= \frac{V^2}{\vec{Z}_L^*}$$

Que. → 2, TF A & B are connected in parallel to a common load $2 + j1.5 \Omega$. Their impedances in z° terms are $z_A = (0.15 + j0.5) \Omega$ & $z_B = (0.1 + j0.6) \Omega$. Their terminal voltages are $E_A = 20 \angle 0^\circ$ & $E_B = 20 \angle 0^\circ$. Find the power o/p & PF of each TF.

Soln →

$$\vec{I}_{sc} = \frac{\vec{E}_A}{\vec{Z}_A} + \frac{\vec{E}_B}{\vec{Z}_B}$$

$$= \frac{20 \angle 0^\circ}{(0.15 + j0.5)} + \frac{20 \angle 0^\circ}{0.1 + j0.6}$$

$$= 732.105 \angle -76.625^\circ$$

$$\frac{1}{\vec{Z}_p} = \frac{1}{\vec{Z}_L} + \frac{1}{\vec{Z}_A} + \frac{1}{\vec{Z}_B}$$

$$(\vec{Z}_p)^{-1} = (\vec{Z}_L)^{-1} + (\vec{Z}_A)^{-1} + (\vec{Z}_B)^{-1}$$

$$\vec{Z}_p = 0.258 \angle 72.85^\circ$$

$$\vec{V} = \vec{I}_{sc} \cdot \vec{Z}_p$$

$$= 189.25 \angle -3.78^\circ \text{ V}$$

$$\vec{I}_A = \frac{\vec{E}_A - \vec{V}}{\vec{Z}_A} + \frac{\vec{E}_B - \vec{V}}{\vec{Z}_B} = 42.21 \angle -38.81^\circ \text{ A}$$

$$\vec{I}_B = \frac{\vec{E}_B - \vec{V}}{\vec{Z}_B} = 33.57 \angle -42.87^\circ \text{ A}$$

$$\vec{S}_A = \vec{V} \cdot \vec{I}_A^*$$

$$= 7.988 \angle 35.03^\circ \text{ kVA}$$

$$\text{i.e. } 6.541 \text{ kW at } \cos 35.03^\circ \text{ lag}$$

$$\text{i.e. } 0.8189 \text{ PF lag}$$

$$\vec{S}_B = \vec{V} \cdot \vec{I}_B^*$$

$$= 6.353 \angle 39.09^\circ \text{ kVA}$$

$$\text{i.e. } 4.931 \text{ kW at } \cos 39.09^\circ \text{ PF lag}$$

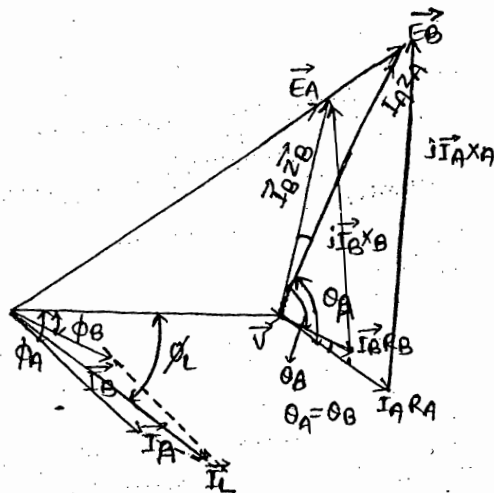
$$\text{i.e. } 0.7762 \text{ PF lag}$$

$$\vec{S}_L = \vec{S}_A + \vec{S}_B$$

$$= 14.332 \angle 36.828^\circ$$

Phasor diagram →

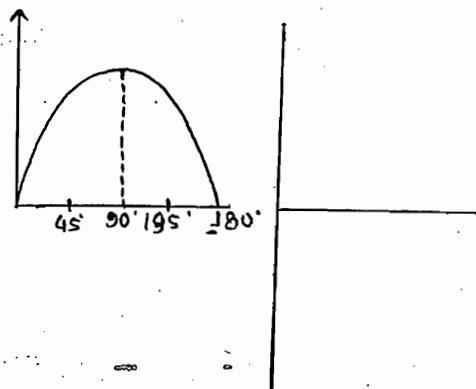
$E_A > E_B$, $\theta_A = \theta_B$, lagging PF load.



* magnetisation Current phenomenon in the TF →

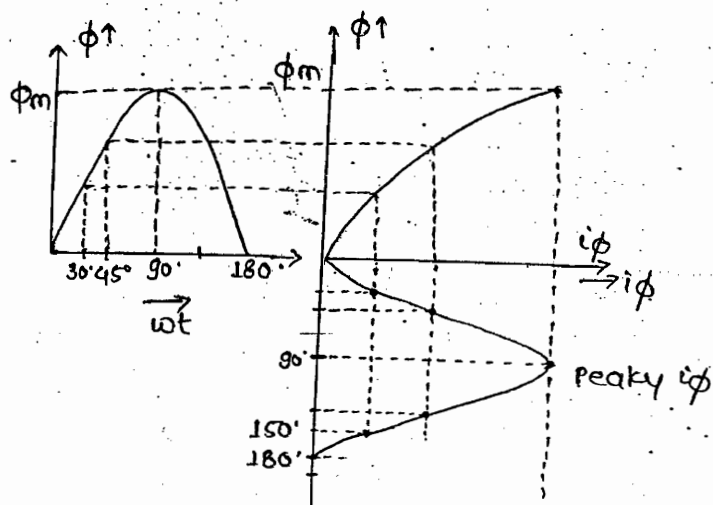
* If the applied vol. to the TF is sinusoidal then the core flux should also be sinusoidal.

* If the magnetisation curve of core material would have been linear then the magnetising current would also have been sinusoidal.



* Model TF are operated with high flux density in the core due to economic reason & this drives the core into deep saturation.

Obviously the magnetisation curve becomes highly non-linear.



* With such a non-linear magnetisation c/s a sinusoidal flux may only be obtained with a peaky magnetising current containing dominant a 3rd harmonic component.

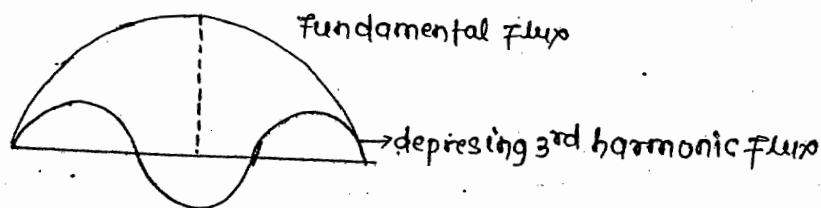
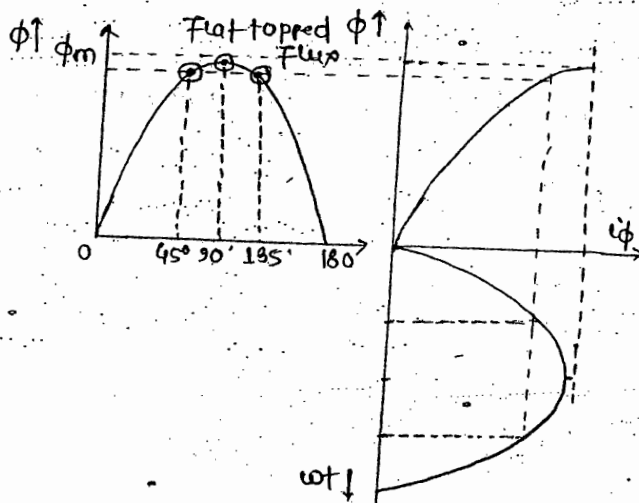
5. The 3rd harmonic component of the magnetising current may flow only if the electric circuit permits.

Thus it can easily flow in 1- ϕ TF of 1- ϕ circuit. However in 3- ϕ TF the 3rd harmonic current of 3 phasors are all in phase; thereby constituting 0 seq. current.

Hence they can flow only in the 1^o star neutral of the TF is connected to the source neutral (or) it can flow in closed Δ .

* In a 1-1 TF there is no path for flow of 3rd harmonic components of magnetising current & therefore the magnetising current remains sinusoidal if higher non-triplank odd harmonics are neglected.

* With a sinusoidal magnetising current the core flux becomes flat topped containing dominant depressing 3rd harmonic flux component.



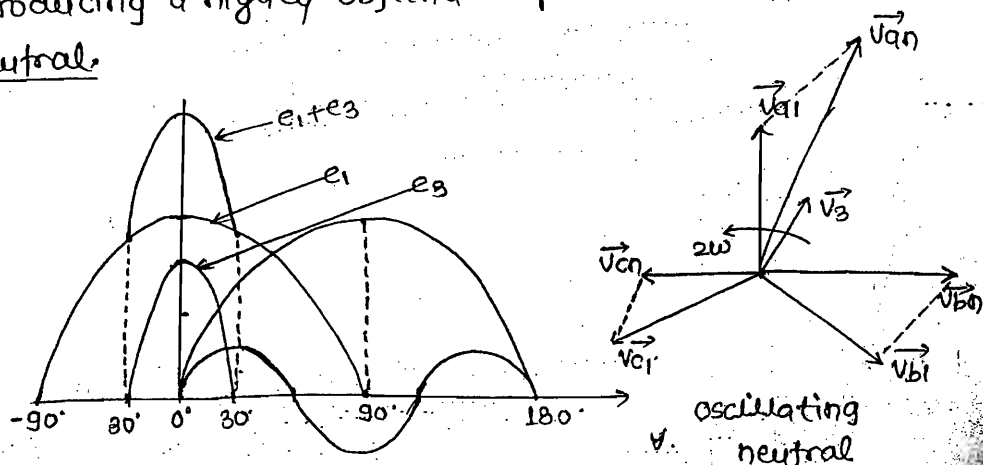
* The 3rd harmonic component of flux may get easily establish only if magnetic ckt permits.

Hence the 3rd harmonic flux components may get established easily in those TF that have independent magnetic ccts such as in 3- ϕ bank of 1- ϕ TF, 3- ϕ shell type TF & 5 limbed (or 4 limbed) core type TF.

* Consequently in magnetically independent TF the 3rd harmonic flux gets strongly established resulting into a peaky induced emf in both wdg's; that in creates high insulation stress in both wdg's in addition to producing a highly objectionable phenomenon called oscillating neutral.

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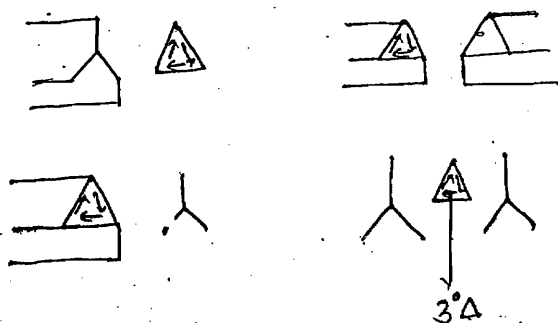
$$\vec{I}_{a3} = \vec{I}_{b3} = \vec{I}_{c3} = \vec{I}_3$$



* The solⁿ to the above prob. is to provide a 3°Δ wdg in a 1-1 TF so that the electric ckt's permits flow of 3rd harmonic components in the closed 3°Δ, that eventually would restore harmonic variation to the core flux.

* The presence of a Δ-wdg provides a low reluctance path for 0-seq. currents & therefore restores harmonic variation (sinusoidal) to the core flux & also facilitates 1-φ loads besides reducing the effect of floating (or) shifting neutral.

It also prevents current choking in 1-1 TF. The presence of Δ also helps in detection of unsymmetrical faults; particularly those involving ~~draw~~ ground.



The problems discussed above relate to those TF that have magnetically independent ckt's.

However a 3-limb core type TF has magnetically interlinked ckt & therefore the 3rd harmonic flux forming a seq. Finds a very high reluctance path through air & tank valves.

Consequently the magnitude of 3rd harmonic flux remains extremely low although it ultimately results in tank valve heating.

Therefore large 3 limb core TF are provided with a cu ring that surrounds the core thereby providing a cu screen for the 3rd harmonic flux & minimized tank valve heating.

Hence a 3-limbed core type TF may be connected in Δ - Δ without a $3^\circ\Delta$. However if the unbalance is expected to exceed then it is recommended to use a $3^\circ\Delta$ wdg.

* A 3 limbed core type TF does not permit easy excess to 3rd harmonic flux & this results into increased 5th & 7th harmonic components in the magnetising current.

These component can't be suppressed by any electric connection. Hence if these harmonics are objectionable then a 4-limbed (or) 5 limbed st. should be provided.

Notes:- The 1st Δ neutral in a Δ - Δ connection is not connected to the gen^r neutral to provide a path for 3rd harmonic component of magnetising current.

This is because the gen^r ϕ vol. itself contain 3rd harmonic vol. & this would appear across the 1st ϕ of TF.

Obviously then the 2nd Δ ϕ would now contain 3rd harmonic vol. component in all its 3- ϕ .

Since the 2nd Δ neutral would be grounded, 3rd harmonics currents would flow in the xⁿ line resulting into objectionable communication interference. Thus the other alternative of providing a $3^\circ\Delta$ wdg is adopted to allow flow of 3rd harmonics of magnetising current.

Miscellaneous →

Phantom → Rated current flowing but not rated load.
loading

* Shell type → High current low voltage (cu surrounded iron)

* Core type → High vol. low current.

Iron surrounded cu.

* Yoke - 20% higher than core
(cross sectional area)

*  area ↑

* Core type - circle wdg (round coil) so max^m area fore

* Stepped core is used in big TF.

* two stepped - crossiform (max^m area) (min^m perimeter)

* Stepping - provides circular shaped (for max^m mechanical strength)

* linear dimensions → k times

kVA rating - k^4 times

losses - k^3 times

surface area - k^2 times

If $k=3$, kVA - 81 & loss - 27 times, area - 9 times

so heat dissipation problem.

so linear dimension ↑ then improve cooling for this

REM → moment of inertia - k^5 times

for small TF - cooling (air natural) (AN)

Big TF - cooling (air forced / air blast)

oil (oil natural air natural) (ONAN)

oil (oil forced - through pump) (OFAN)
air natural radiator size ↑

oil forced air forced (OFAF)

* Tap changer $\rightarrow$ ON load (costly) cooling must be good
 $\rightarrow$ off load (cheap)

Tap changer $\rightarrow$ HV side
 (100:1)

If one turn change in LV then 100v in HV.

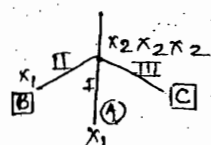
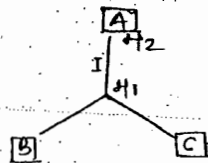
Taps are used in center because of axial force are balanced in

* Core type - circular coil
 Shell type - Rectangular/
 $\rightarrow$ Helical wdg (HV)
 $\rightarrow$ disc (HV) invariably large TF

Spiral wdg - 2000 100A $\uparrow$

Cross over coil - 20A (max^m) low capacity HV coil

(22/17)

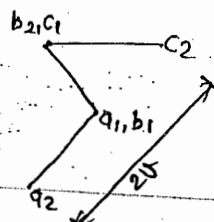
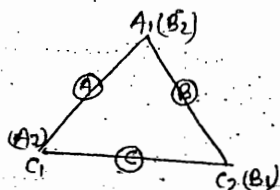


(23/17)

ϕ is +ve for lagging PF.

(24/17)

(32/19)



(35/19)

